Corrected CEAC finite-model lane and v90 specialization ledger

Date: 2026-11-10. Reference: `doc/ceac_v90_accessibility_release`, Section 3 and Appendix D. The reference manuscript, released algorithms, historical runs and figures were not edited. The maintained implementation is `code/ceac`.

This report records an implementation, not a retroactive reinterpretation of the old plots. Historical FrozenLake's fixed-route agent is still what produced those plots. New cells identify the corrected source, configuration, model family, policy class, estimator and prior explicitly.

Scope and source ownership

Both benchmarks now use one shared episode loop with a softmax actor, reward critic, row-indexed evidence critics, immutable policy snapshots, full policy-labeled trajectory replay, exact bounded design solves, the finite joint target/calibration program, global separation, and cumulative execution debt. The old environment dynamics were copied verbatim into `code/ceac/environments` as the starting maintained kernels. No old learner or runner is imported. The release remains a frozen reference.

This first lane uses the known finite candidate transition/reward laws already assumed by these benchmarks. It does **not** claim an unknown-dynamics, simulator-free implementation. Candidate-model planning calls are separated from real-environment trajectory acquisition, and do not enter the actor's training buffer as if they were self-play. Runtime is recorded; planning calls are not counted as physical environment episodes.

1. Public observations, model class and policy class

The public class has modes 0 and 1. The realized simulator is owned by the runner. Its `mode`, `optimal_policy`, `shore_label`, configuration and `info` dictionaries are not sent to CEAC or IDS. The learner receives only a separately constructed model-family interface and observations/actions/rewards. Only the explicit Lai–Graves arm receives the realized mode.

All three methods have the same policy space: mask-respecting finite-horizon Markov policies on the encoded observation, which includes elapsed time and observable signals. The banks are computational active sets, not restrictions on what the corrected oracle or IDS may search. Stochastic actor snapshots are valid columns in the convex hull of complete deterministic Markov policies.

The exact transition kernel is enumerated from actual `step()` outcomes. It checks common positive support, normalized laws, one random event per step, and increasing elapsed time. A changed kernel that violates those assumptions raises an error instead of silently producing a certificate.

2. Full-policy separation without revealing the model

For these two specific structures, the implementation verifies a policy-independent state-occupancy ratio

$$d_m^\pi(s) = w_m(s) d_0^\pi(s), \qquad w_m(s_0) = 1.$$

Every nonterminal edge satisfies

$$w_m(s') = w_m(s) \frac{P_m(s' \mid s, a)}{P_0(s' \mid s, a)}$$

for every predecessor/action reaching that state, marginalizing emitted rewards for state occupancy. All paths must give the same ratio. For arbitrary coefficients c_m, b_{mj}, a *single shared* policy maximizing

$$\sum_m c_m J_m(\pi) + \sum_{m,j} b_{mj} I^m(\pi; j)$$

can therefore be found by backward induction under P_0, using stage reward

$$g(s,a) = \sum_m w_m(s) \left[c_m r_m(s,a) + \sum_j b_{mj} \kappa_{mj}(s,a) \right].$$

This is not a separate policy maximum in each hidden model. The same action is selected at each public state. The check is benchmark-specific: a general MDP with path-dependent occupancy ratios needs another structured response oracle. The implementation fails closed in that case.

For IDS, a further check verifies that any reward evidence absent from the next encoded state has **no future informative descendant**. This permits state-based predictive beliefs in the trajectory mutual-information calculation. DeepSea's challenge reward is the relevant example: its information is counted at that step, and no later diagnostic can depend on forgotten evidence. This structural condition is checked rather than assumed for arbitrary environments.

3. Likelihood confidence and exact candidate-model arrays

The previous empirical information LCB is removed from the maintained lane. Replay still stores all returns and per-model transition log likelihoods under the immutable behavior-policy label. These empirical statistics are diagnostics; they are not mislabeled confidence bounds.

Candidate-model values $J_m(\pi)$ and trajectory KLs $I^m(\pi; j)$ are instead computed exactly by finite-horizon evaluation. Sampling error in these arrays is zero conditional on the declared model family. The unknown quantity remains which candidate generated the actual trajectories.

For equal initial log scores,

$$S_t(m) = \sum_{k<t} \log p_m^{\pi_k}(\tau_k), \quad \mathcal{C}_t = \{m : \max_j S_t(j) - S_t(m) \le \log((|\mathcal{M}| - 1)/\delta)\}.$$

For any true candidate m, each likelihood ratio $\exp(S_t(j) - S_t(m))$ is a nonnegative martingale under predictable adaptive policy selection. Ville's inequality and a union bound over $j \neq m$ give

$$\Pr_m\{\exists t : m \notin \mathcal{C}_t\} \le \delta.$$

The confidence allocation is divided across contexts when more than one is requested. It is a per-run statement, not a simultaneous statement across all independent seeds/methods in a study. The uniform likelihood mixture $\xi_t = \mathrm{softmax}(S_t)$ supplies the randomized model estimate for the finite target/calibration LP. This uniform initialization does not import the IDS comparator's deliberately excluded prior support.

Rows are rebuilt for every (m, j) with $m \in \mathcal{C}_t$ and $j \in \mathcal{M}_{\mathrm{alt}}(m)$. Rejected references do not impose permanent rows, while their stored trajectories and previously issued quotas remain.

To make the cost bound conservative before model resolution while retaining a linear full-policy response, define

$$U_t^\star = \max_{m \in \mathcal{C}_t} J_m^\star, \qquad L_t(\pi) = \min_{m \in \mathcal{C}_t} J_m^\star - \sum_{m \in \mathcal{C}_t} \Delta_m(\pi).$$

On the plausible-model event, $L_t(\pi) \le J_{m_\star}(\pi)$, so $\overline{\Delta}_t = U_t^\star - L_t$ is a valid upper bound on true regret. For a singleton it is the exact regret. This deliberately loose model-robust bound replaces the paper's sampled return LCB in this exact-evaluation lane. It is a disclosed specialization, not a new empirical concentration claim.

4. Bounded master, global oracle and column admission

The primal and dual are v90's bounded programs:

$$\min_{\eta \ge 0} \overline{\Delta}^T \eta \quad \text{s.t. } K\eta \ge \mathbf{1}, \quad \mathbf{1}^T\eta \le B_{\max},$$

$$\max_{\beta,\lambda \ge 0} \mathbf{1}^T\beta - B_{\max}\lambda \quad \text{s.t. } K^T\beta \le \overline{\Delta} + \lambda\mathbf{1}.$$

SciPy/HiGHS solves the LP; a globally maximizing common-policy response checks all omitted columns. Any negative-reduced-cost separator is inserted and the master is solved again. The implementation records row coverage, primal/dual gap, mass price, evidence prices and global column residual. A failed solver or unclosed certificate raises an error; no fixed-bank fallback is retained.

The seed-mixture coverage bound is checked explicitly. The default $B_{\max} = 20$ contains the displayed certified optimizers. The normal-mode full-policy coefficients are

- DeepSea: $0.015/1.21422073877 = 0.0123536022084$.
- FrozenLake: $0.042/1.21422073877 = 0.0345900861836$.

In shifted mode both coefficients are zero because the optimal policy itself is informative. FrozenLake's normal-mode separator discovers the 13-step survey-to-inner-dock route omitted by the released six-route bank.

Actor snapshots are evaluated with the same reduced-cost upper bound before admission. Because an exact separator already closes the full policy residual, a later actor proposal need not add another column under unchanged prices. Column discovery in this lane therefore cannot be attributed solely to the actor. Exact global response is an explicit part of tabular v90.

The corrected oracle uses expected design counts with the same cumulative scale as CEAC. It does not use the released DeepSea scheduler's realized-LLR stopping rule. Its finite-sample execution and asymptotic coefficient are reported separately.

5. Target/calibration program and finite schedule choices

For each candidate good set $G \subseteq \{0,1\}$, the implementation solves Appendix D's joint LP in (λ, ω, t, v), including the zero-GLC models. Target acceptability uses the certified per-model coefficients and $\delta_\epsilon = \epsilon/6$. Calibration uses

$$\overline{I}_t(\pi; m) = \sum_j \xi_t(j) I^j(\pi; m).$$

Both target and calibration column searches use the shared-policy oracle above. A restricted infeasible LP is not automatically called globally infeasible: phase-I slack minimization also receives full-policy separation. Only a positive globally certified phase-I slack rules out that candidate set. All candidate solutions are refreshed against the same final bank.

If the all-model good set is feasible, calibration is unnecessary; its otherwise unconstrained epigraph variable is fixed to zero. Other feasible sets are ranked by their calibration value, with larger good sets preferred on

exact ties. This makes the unbounded all-good edge case explicit rather than interpreting an unbounded solver result as failure.

The mixture follows Appendix D:

$$p_t = q\lambda_t + (1-q)\omega_t, \quad q = \frac{4n_{\max} + \epsilon g_{\mathcal{M}}}{4n_{\max} + 2\epsilon g_{\mathcal{M}}}.$$

For the correctness slate, $\epsilon = 0.25$, $n_{\max} = 20$, refresh interval 8 episodes, and snapshot interval 4 episodes are explicit configuration. The normalization contains the checked finite designs; it is **not** presented as a derivation of every uniform-regularity constant in the imported AE2 theorem. At an unresolved refresh, if $\mathcal{C}_t \not\subseteq G$, one draw from p_t requests a fresh compatible certificate trajectory. Pending compatible requests coalesce. The ordinary CEAC selector then executes that request through the same episode loop as design debt and actor episodes.

The supplied draft does not completely specify how many estimator-oracle calls are interleaved with each physical episode or how its imported estimator and stopping rule map to that cadence. The schedule above is consequently an explicit implementation choice. A coefficient-one theorem for this exact schedule is **not** claimed. An asymptotic experiment also needs the declared horizon-dependent tolerance schedule and justified class constants; a fixed small validation slate cannot establish those claims.

6. Actor/critic update and its precise representation change

For $\pi_\theta(a|s) = \mathrm{softmax}(\theta(s,\cdot)/T)_a$, the score is

$$\nabla_{\theta(s,\cdot)} \log \pi_\theta(a|s) = (\mathbf{e}_a - \pi_\theta(\cdot|s))/T.$$

The actor gradient includes this factor. Each episode freezes the behavior policy and evidence prices. All TD errors are computed before any update; the actor is updated first, then the critics. Only fresh actor episodes update either actor or critic. Stored replay and frozen diagnostic episodes never supply off-policy actor gradients.

Instead of one scalar evidence table reused across prices/references, keep unpriced row critics V_{mj} with immediate reward $\ell_{mj} = \log(p_m/p_j)$, and form

$$V_\beta^I = \sum_{m,j} \beta_{mj} V_{mj}, \qquad \delta_\beta^I = \sum_{m,j} \beta_{mj}[\ell_{mj} + V_{mj}(s') - V_{mj}(s)].$$

Each row takes its own on-policy TD update. At a fixed policy the Bellman operator is linear in reward, so the combined fixed point is exactly the priced-evidence value. At $\beta = 0$ the evidence contribution vanishes identically even if the unpriced tables contain old values.

This basis-critic update is a representation-level refinement of the draft's single parameterized evidence-critic update, not identical parameter dynamics. It avoids contamination from old prices or reversed reference signs. It does not establish critic convergence from finite data by itself.

The validation initializes the actor from the optimal policy of the **uniform candidate-model mixture**, with 10% valid-action uniform mixing. This uses public model knowledge, not the realized mode. `--actor-init uniform` provides a control. Reward and evidence value tables start at zero. The actor and critic step sizes decay with actor-update count, with exponents 0.75 and 0.6, respectively. No hand-coded environment-specific route preference is installed.

7. Persistent quotas, statistical debt and evidence

At refresh, use the stated slowly inflated logarithmic scale b_r, and add

$$Q_{r,i} = Q_{r-1,i} + \frac{b_r - b_{r-1}}{\min(1, \min_j (K\eta_r)_j)} \eta_{r,i}.$$

The denominator is the certified row-shortfall repair. Old quotas are never replaced by $b_r \eta_r$. Statistical requirements and design quotas have separate counters; the selector serves statistical debt, then the largest design debt, then the actor. One actual trajectory increments the immutable policy's physical count once and the compatible certificate count once. It can satisfy both debts without being purchased twice. Replaying the same episode ID twice is an error. The one-episode rounding guarantee is checked after debt is cleared; it is not asserted during an unfinished debt block.

Every episode records context, immutable policy ID, behavior type, states, actions, rewards, termination and per-model transition log likelihoods. Frozen behavior probabilities are persisted even for unadmitted actor versions. Source/config/runtime/seed/prior identities and artifact hashes prevent count-only resumption from silently reusing old science. Method RNGs are keyed by stable identifiers rather than method-list position.

8. IDS comparator and support-exclusion controls

The corrected comparator updates exact finite-model likelihood weights and minimizes the information ratio over policy mixtures using an analytic pair-support solve. It has no rounded posterior cache or mixture grid. Independent rescaling of regret and information protects tiny posterior tails from absolute thresholds; underflow of a still-positive supported probability raises an error. A common likelihood factor leaves the posterior unchanged exactly.

For $D = a + bx$, $I = c + dx$, the interior stationary mixture satisfies

$$x = (da - 2bc)/(bd),$$

when it lies in $(0, 1)$; endpoints are also checked. A full-policy conditional gradient certificate searches for the maximum of $\overline{J}(\pi) + (D/2I)\,\mathrm{MI}(\pi)$. If it improves the restricted solution, that policy is added and IDS is solved again. The final residual is scaled to posterior expected regret, not to an arbitrary absolute exploration floor. The binary models have distinct optimal policies, so model MI and optimal-policy MI coincide here; a larger family must revisit that identity.

Full-support IDS remains the primary control. `ids_rare` has prior 0.001 on the shifted mode; `ids_zero` has prior zero and is explicitly labeled misspecified. With zero shifted support, the confounded old route yields exactly 0.04 regret per episode in DeepSea and 0.208 in FrozenLake when mode 1 is true. This is a support-exclusion experiment, not a proof that all Bayesian IDS variants fail.

9. Acceptance checks and limits of the fresh slate

The regression suite checks all 22 released policy/model return/KL pairs, both full-policy oracle coefficients and modes, global weighted-policy responses, robust costs, target/calibration feasibility and phase-I certificates, finite-difference actor scores, the zero-price specialization, absence of frozen-policy actor updates, confidence first-crossing probabilities, immutable snapshots, single credit, cumulative quotas, tiny IDS priors, exact trajectory MI, absorbing zero support, context-specific interfaces, stable seeds, and verified persistence/incompatible resumption.

The short validation slate is stored in `code/runs/ceac_v90_validation_20261110`, with a separately rendered `report.md`. It uses 256 episodes, both candidate modes, three seeds, and five methods in each of the two environments. It is an integration/control slate, not a parameter sweep. Actor progress and model/design certificates are reported separately.

The current contexts are stationary. A later unannounced within-context shift will require a declared monitoring/restart or nonstationary likelihood model, segmentation of historical replay, and matched hazard-aware IDS controls. Old stationary confidence cannot simply be carried through such a shift. Structural support-hole constructions must likewise stay inside CEAC's declared candidate family if its realizability-based guarantees are invoked.

10. Earlier actor-only validation and diagnosis (archived)

The completed slate contains **60 runs and 15,360 real-environment episodes**. All run artifacts verify, and a full resume skips the completed cells by exact identity. All **22 regression tests** pass. Across **384 CEAC refreshes**, the maximum recorded primal/dual gap and global column residual are zero at the reported numerical precision. All 12 CEAC cells finish with the correct singleton model set, and all 12 actors improve their expected return. The numerical validation summary is `audits/ceac_v90_validation_20261110.json`.

That does **not** establish CEAC performance superiority. Mean final actor returns in shifted mode are 0.77984 for DeepSea (optimum 1.024) and 0.89056 for FrozenLake (optimum 1.127). Full-support IDS reaches the optimal route in the short slate. Zero-support IDS remains exactly on the confounded old route, incurring 10.24 and 53.248 cumulative pseudo-regret respectively over 256 episodes. Positive rare-prior IDS sometimes recovers, illustrating why rare support must not be conflated with missing support.

A post-hoc check of the saved actor tables separates residual random actions from route preference. Taking the greedy action of each final shifted-mode actor yields return 0.984 in DeepSea and 1.0538 in FrozenLake, both below their full optima. Thus the remaining gap is not explained solely by stochastic execution. These greedified policies are **diagnostics only**; they were not substituted into the training run or used to inflate its reported performance.

The correct shifted model was first isolated between episodes 3 and 16 in DeepSea and 6 and 8 in FrozenLake (zero-based indices). The final evidence prices are zero in that mode, as they should be when the optimal diagnostic policy has zero regret. Frozen diagnostic trajectories do not train the actor under the specified on-policy rule. Consequently correct model identification and a correct experimental design do not instantly teach the actor the optimal deployment route. The primitive actor still has to learn it through its own rollouts. Longer-time actor convergence and the coupling between certified model knowledge and deployment must be checked before using this slate as a CEAC-recovery claim or launching a larger shift sweep. No planner policy has been silently substituted for the actor to conceal that gap.

11. Deployment correction and expanded comparator slate

The deployment gap above exposed one further interface omission. Appendix D's `finite-deployment-response` defines an estimator-averaged deployment policy; Section 4's finite allocation-estimation lemma explicitly requires an exact deployment response and separately charges pre-certification deployment regret. Appendix G, Step N4, applies the reward-only certificate at $\beta = 0$ to make deployment optimal without assuming actor SGD has found a global optimum. Section 3's episode selector, however, says to use the actor whenever no debt is owed and does not explicitly connect that deployment response to execution.

The previous maintained runner followed that actor-only branch, leaving a learned model and a certified experimental design disconnected from deployment. The corrected default makes the connection explicit;

`ceac_actor` retains the actor-only branch as an ablation with the same seeds, design logic and budget. The v90 manuscript is unchanged.

Let ξ_t^C be the normalized likelihood estimate restricted to the current plausible model set. Compute a public-model reward response

$$\widehat{\pi}_t \in \arg\max_{\pi} \sum_{m \in \mathcal{C}_t} \xi_t^C(m) J_m(\pi), \qquad \epsilon_t^{\text{deploy}} = \max_{m \in \mathcal{C}_t} \{J_m^\star - J_m(\widehat{\pi}_t)\}.$$

When $\epsilon_t^{\text{deploy}} \leq 10^{-10}$, deployment is certified for every plausible model. On the anytime model-confidence event, it is therefore equally accurate for the true model. The proposal uses no realized-mode label. When the confidence set changes, both design and deployment certificates are refreshed; a certificate is invalid outside its certified model set.

Execution serves statistical debt, then design debt, then certified deployment when available, and otherwise the current actor. Frozen certified-deployment episodes are labeled as such and enter likelihood/replay and execution counts; they never train the actor off policy. Exact deployment therefore resolves a policy-execution gap, **not stochastic actor convergence**. In these small finite-model cases a certificate can precede the first free actor episode, so some cells have zero actor updates. Those cells validate the planner-assisted finite implementation, not actor discovery. The report preserves both actor and deployment performance, counts and policy identities.

Restricting the proposal weights to the confidence set and specifying this execution priority are explicit implementation choices that resolve the draft's missing bridge. The all-model reward-gap certificate is independent of how its proposal is produced. The standing caveats about normalization, estimator cadence, stationary replay and the coefficient-one theorem still apply.

Comparators and what is actually matched

The expanded slate contains the corrected CEAC, the actor-only ablation, fixed offline Lai–Graves, model-confidence optimism, coherent policy-UCB, a policy-level OIR/IDAC adaptation, uniform coverage, and full/rare/zero-support IDS. Each gets 256 episodes for each environment and mode, with three paired seeds. Only the offline oracle receives the realized mode; its allocation is solved once and is never updated using observed rewards or likelihoods. Its count schedule remains the declared logarithmic cumulative schedule.

The model-confidence baseline optimizes over its anytime likelihood-confidence set using the same declared model class and full policy responses as CEAC/IDS. This is a strong control; confidence-driven exploration is not synonymous with a poorly tuned stepwise bonus.

The three finite-bank controls share one **enriched public bank**, constructed without knowing the realized mode. It contains both model-optimal policies and the cheaper survey-to-inner FrozenLake route. Equivalent policies on reachable trajectories are deduplicated so arbitrary off-path action assignments do not inflate the number of bandit arms. These controls do not lose because their bank omits the necessary deployment or diagnostic route. They nevertheless remain bank-based algorithms and do not share CEAC/IDS's unrestricted response oracle.

Policy-UCB uses the entire attainable episodic return range R, found by an all-outcome finite-horizon recursion. Its radius for arm i after n samples is

$$r_n = R\sqrt{\frac{\log(L\pi^2 n^2/(3\delta))}{2n}},$$

where L is the bank size. Hoeffding plus a union bound over arms and counts spends at most δ. The old hard-coded bonus multiplier 0.18 is not reused as if it were a calibrated confidence bound.

OIR/IDAC uses the released experiment's finite-policy occupancy-ratio principle, with the explicit objective

$$F(q) = \frac{\sum_i q_i(1.25 - \widehat{J}_i)}{0.25 + H(\sum_i q_i \widehat{d}_i)}.$$

Here d_i is the expected distribution of a uniformly sampled pre-action state within one episode, including observable survey/noise symbols encoded in that state. Each observed trajectory contributes unit normalized occupancy mass. The total occupancy pseudocount is 0.25 per policy, spread across the alphabet; it is not 0.25 times the number of encoded states. Each policy is sampled once before ratio updates. One exact softmax-logit gradient step with step size 0.2 follows each episode, using empirical returns and occupancies only. The gradient is checked against finite differences. The cost reference exceeds every possible episodic return in both default benchmarks.

This is transparently a **finite episodic policy-level adaptation**, not a claim to reproduce the original infinite-horizon OIR/IDAC implementation. Uniform coverage and policy-UCB use the identical public bank. These comparisons do not establish superiority over every occupancy objective, confidence method or neural function approximator.

Reporting and math rendering

The new immutable evidence is in `runs/ceac_deployment_comparison_20261110`; earlier cells remain untouched. `ceac.report` renders expected-return, cumulative-pseudo-regret and cumulative model-KL traces, plus deployment/prior ablations, from verified records. Bands show the observed seed range and are not mislabeled confidence intervals. Primitive interaction counts are reported alongside episode budgets. Runtime source identity excludes the report/figure modules, whose hashes are recorded separately in the rendering audit. Cosmetic edits therefore do not invalidate scientific experiment caches. The current deployment renderer refuses to reinterpret the archived actor-only slate as certified CEAC.

Markdown does not compile LaTeX itself. The previous reports used `\(...\)` and `\[...\]`, which are not consistently recognized by Markdown math extensions. The mathematical reports now use `$...$` and `$$...$$` outside code blocks. A math-aware Pandoc parse verifies recognition, and a compiled PDF is provided as a viewer-independent alternative. No manuscript TeX sources were modified.

12. Completed deployment comparison

The fresh comparison completed **120 runs and 30,720 environment episodes**. All **31 regression tests** pass, including certificate revocation, live 64-episode gap closure, fixed-oracle allocation, finite-difference OIR gradients, confidence-range calibration and trajectory-only comparator isolation. Cached-prefix checks also certify the correct reward response in all 12 earlier CEAC cells once their model-confidence set resolves.

In every one of the 12 fresh CEAC cells, last-window expected return matches the paired fixed offline oracle to numerical tolerance. In shifted mode this is **1.024 for DeepSea and 1.127 for FrozenLake**. Normal-mode last-window means are 0.983531 and 0.917687, including the scheduled diagnostic episodes; the separate certified deployment responses are optimal at 0.984 and 0.919.

Mean cumulative pseudo-regret after 256 episodes in shifted mode:

Method	DeepSea	FrozenLake
CEAC with certified deployment	2.4312	2.4368
CEAC actor-only ablation	72.5510	60.3118

Method	DeepSea	FrozenLake
Fixed offline Lai–Graves	0	0
Model-confidence optimism	0	0
Policy-UCB on enriched bank	12.8967	40.8458
OIR/IDAC policy adaptation	47.3577	63.1475
Uniform coverage	77.0267	64.0565
IDS with full support	0.0280	0.0732
IDS with rare prior 0.001	6.2937	30.6189
IDS with zero shifted support	10.2400	53.2480

These results close the diagnosed deployment gap and reproduce the intended support-exclusion failure. They **do not show CEAC beating full-support IDS or model-confidence optimism** on this simple binary family. Those controls solve the problem with less startup cost. The plots retain them, alongside the finite-budget advantages over the tested coverage/ratio and rare/zero-prior arms. No confidence level, seed, model support or reward was changed to create a more favorable ranking.

The comparison report includes PNG and vector-PDF figures. The mathematical ledger and comparison report also compile to PDF, and the run contains a hash-checked archive of its runtime source and regression tests. No historical run, published figure or manuscript source was overwritten.